

## PATHUMTHANI, Thailand

WMO: 484190

Lat: 14.1162N

Lon: 100.6203E

Elev: 7

StdP: 101.24

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 19.2       | 20.8 | 12.5                          | 9.0 | 21.8 | 14.5 | 10.3 | 23.4 | 5.3                   | 27.5 | 4.5 | 27.8 | 1.5                   | 10   | 0.369 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 8.9                    | 37.4            | 27.7 | 36.5 | 27.4 | 35.7 | 27.2 | 29.3                | 35.0 | 28.7 | 33.9 | 28.2 | 33.1 | 2.8                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 28.0                            | 24.1 | 32.6 | 27.5 | 23.4 | 31.6 | 27.0 | 22.8 | 30.8 | 96.5          | 35.2 | 93.4 | 33.9 | 90.9 | 33.2 | 35.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 5.6               | 5.0  | 4.3 | 16.5                       | 38.2 | 2.0                | 1.2 | 15.1                                               | 39.0 | 13.9       | 39.7 | 12.8       | 40.3 | 11.4       | 41.2 |    |    |
|                   |      |     | 13.8                       | 30.1 | 2.1                | 2.1 | 12.3                                               | 31.6 | 11.0       | 32.9 | 9.8        | 34.1 | 8.2        | 35.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 29.3   | 27.0  | 28.7  | 30.0  | 31.1  | 31.0  | 30.2  | 29.7  | 29.6  | 29.3  | 29.3  | 28.8  | 27.3  |      |
|                                                                               | DBStd                                                                         | 2.00   | 2.13  | 1.53  | 1.88  | 1.69  | 1.64  | 1.12  | 1.25  | 1.06  | 1.14  | 1.29  | 1.57  | 2.18  |      |
|                                                                               | HDD10.0                                                                       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |      |
|                                                                               | HDD18.3                                                                       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |      |
|                                                                               | CDD10.0                                                                       | 7048   | 526   | 523   | 619   | 632   | 649   | 606   | 611   | 608   | 578   | 597   | 563   | 535   |      |
|                                                                               | CDD18.3                                                                       | 4006   | 267   | 289   | 361   | 382   | 391   | 356   | 352   | 350   | 328   | 339   | 313   | 277   |      |
|                                                                               | CDH23.3                                                                       | 49538  | 2849  | 3516  | 4743  | 5273  | 5263  | 4623  | 4388  | 4285  | 3889  | 4043  | 3679  | 2986  |      |
|                                                                               | CDH26.7                                                                       | 23418  | 1151  | 1643  | 2456  | 2972  | 2841  | 2277  | 2013  | 1920  | 1641  | 1724  | 1573  | 1206  |      |
| Wind                                                                          | WSAvg                                                                         | 1.6    | 1.2   | 1.4   | 1.8   | 1.8   | 1.9   | 2.0   | 2.0   | 2.0   | 1.6   | 1.1   | 1.3   | 1.4   |      |
| Precipitation                                                                 | PrecAvg                                                                       | 1207   | 10    | 9     | 37    | 76    | 154   | 131   | 134   | 157   | 273   | 182   | 39    | 6     |      |
|                                                                               | PrecMax                                                                       | 1461   | 52    | 34    | 124   | 198   | 308   | 225   | 255   | 251   | 388   | 437   | 145   | 43    |      |
|                                                                               | PrecMin                                                                       | 885    | 0     | 0     | 2     | 8     | 37    | 28    | 60    | 85    | 168   | 40    | 0     | 0     |      |
|                                                                               | PrecStd                                                                       | 162    | 12    | 9     | 37    | 50    | 61    | 43    | 48    | 44    | 66    | 72    | 39    | 10    |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 34.7  | 35.8  | 37.5  | 38.7  | 38.4  | 36.7  | 36.0  | 35.6  | 35.4  | 35.7  | 35.3  | 35.2  |      |
|                                                                               |                                                                               | MCWB   | 25.7  | 26.8  | 26.9  | 27.8  | 28.5  | 27.7  | 27.7  | 26.9  | 27.3  | 27.4  | 26.7  | 26.6  |      |
|                                                                               | 2%                                                                            | DB     | 33.6  | 34.8  | 36.3  | 37.6  | 37.1  | 35.5  | 34.8  | 34.4  | 34.4  | 34.4  | 34.3  | 34.1  |      |
|                                                                               |                                                                               | MCWB   | 25.0  | 25.9  | 26.7  | 27.8  | 27.8  | 27.5  | 26.9  | 26.6  | 27.0  | 27.0  | 26.4  | 25.9  |      |
|                                                                               | 5%                                                                            | DB     | 32.6  | 34.1  | 35.4  | 36.7  | 36.0  | 34.5  | 33.8  | 33.6  | 33.4  | 33.6  | 33.5  | 33.0  |      |
|                                                                               |                                                                               | MCWB   | 24.7  | 25.7  | 26.6  | 27.5  | 27.6  | 27.2  | 26.7  | 26.5  | 26.8  | 26.8  | 26.2  | 25.4  |      |
|                                                                               | 10%                                                                           | DB     | 31.7  | 33.1  | 34.5  | 35.6  | 34.9  | 33.5  | 32.9  | 32.8  | 32.6  | 32.7  | 32.6  | 31.8  |      |
|                                                                               |                                                                               | MCWB   | 24.4  | 25.3  | 26.5  | 27.3  | 27.4  | 26.8  | 26.5  | 26.3  | 26.6  | 26.7  | 25.9  | 24.7  |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 27.2  | 28.2  | 28.9  | 30.5  | 29.9  | 29.4  | 28.9  | 28.1  | 28.2  | 28.3  | 28.0  | 27.5 |
|                                                                               |                                                                               |        | MCDB  | 32.1  | 34.4  | 35.3  | 36.9  | 36.5  | 34.7  | 34.3  | 32.3  | 33.3  | 33.3  | 33.1  | 33.5 |
|                                                                               |                                                                               | 2%     | WB    | 26.6  | 27.2  | 28.1  | 29.6  | 29.1  | 28.6  | 28.1  | 27.5  | 27.7  | 27.7  | 27.4  | 26.7 |
|                                                                               |                                                                               |        | MCDB  | 31.4  | 32.8  | 33.9  | 35.5  | 34.8  | 33.2  | 32.4  | 31.8  | 32.5  | 32.4  | 32.3  | 32.2 |
| 5%                                                                            |                                                                               | WB     | 26.0  | 26.7  | 27.5  | 28.6  | 28.5  | 27.9  | 27.3  | 27.1  | 27.3  | 27.4  | 27.0  | 26.2  |      |
|                                                                               |                                                                               | MCDB   | 30.6  | 31.9  | 33.0  | 33.7  | 33.6  | 31.9  | 31.5  | 31.5  | 31.7  | 31.8  | 31.7  | 31.2  |      |
| 10%                                                                           |                                                                               | WB     | 25.5  | 26.2  | 27.0  | 27.9  | 28.0  | 27.5  | 26.9  | 26.7  | 26.9  | 27.1  | 26.6  | 25.6  |      |
|                                                                               |                                                                               | MCDB   | 29.7  | 31.1  | 32.3  | 33.0  | 32.7  | 31.4  | 31.0  | 31.0  | 31.0  | 31.2  | 30.8  | 30.1  |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 9.3   | 9.4   | 9.1   | 8.9   | 7.8   | 7.0   | 6.5   | 6.8   | 6.7   | 6.8   | 7.4   | 8.6   |      |
|                                                                               |                                                                               | MCDBR  | 10.0  | 10.2  | 10.3  | 9.9   | 8.9   | 8.2   | 7.7   | 7.7   | 7.9   | 8.0   | 8.5   | 9.3   |      |
|                                                                               |                                                                               | MCWBR  | 3.3   | 2.9   | 2.8   | 2.9   | 2.6   | 2.4   | 2.3   | 2.2   | 2.5   | 2.4   | 2.6   | 3.1   |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 8.7   | 9.3   | 9.6   | 9.5   | 8.5   | 7.6   | 7.1   | 7.2   | 7.3   | 7.3   | 7.6   | 8.7   |      |
|                                                                               |                                                                               | MCWBR  | 3.2   | 3.0   | 3.1   | 3.5   | 3.1   | 2.8   | 2.6   | 2.5   | 2.6   | 2.5   | 2.5   | 3.1   |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          |        | 0.540 | 0.616 | 0.651 | 0.589 | 0.507 | 0.467 | 0.458 | 0.459 | 0.487 | 0.576 | 0.518 | 0.506 |      |
|                                                                               | taud                                                                          |        | 1.954 | 1.801 | 1.753 | 1.937 | 2.220 | 2.330 | 2.358 | 2.341 | 2.254 | 1.964 | 2.100 | 2.075 |      |
|                                                                               | Ebn at Noon                                                                   |        | 758   | 719   | 707   | 751   | 803   | 826   | 836   | 844   | 825   | 742   | 773   | 777   |      |
|                                                                               | Edh at Noon                                                                   |        | 181   | 219   | 235   | 195   | 145   | 128   | 125   | 129   | 140   | 184   | 155   | 157   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        |        | 4.99  | 5.40  | 5.60  | 6.02  | 5.57  | 5.30  | 5.03  | 5.04  | 4.88  | 4.82  | 4.93  | 4.85  |      |
|                                                                               | RadStd                                                                        |        | 0.29  | 0.22  | 0.40  | 0.34  | 0.39  | 0.28  | 0.24  | 0.32  | 0.26  | 0.27  | 0.31  | 0.22  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (33 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page